

Predicting Hospital Performance in California: A Multi-Method Analytics Approach Using Linear and Logistic Regression

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Abstract

This study applies a full-stack analytics pipeline to 120 California hospitals with empirically computed statistics. Linear regression yielded $R^2 = -0.24$ (CV $R^2 = -0.15$), indicating the available predictors explain negligible variance in Performance Rating. Logistic regression achieved 56.7% accuracy and AUC = 0.638. North/West regions and General hospital type were the strongest positive predictors; adverse event rate was the strongest negative predictor. All code: github.com/PatelSahil-HDS.

Keywords: hospital performance, regression, healthcare analytics, scikit-learn, ggplot2, BigQuery ML, Saint Louis University

1. Introduction

Hospital performance ratings serve as composite proxies for care quality, patient safety, and clinical outcomes (Jha et al., 2012). This paper addresses two research questions:

RQ1: Which hospital-level characteristics are most strongly associated with overall performance rating?

RQ2: To what extent can logistic regression classify hospitals as high performers (Rating ≥ 75)?

Complete codebase: github.com/PatelSahil-HDS.

2. Literature Review

2.1 Hospital Performance Measurement

Teaching hospitals exhibit superior outcomes for complex conditions (Burke et al., 2020). Specialty hospitals show mixed evidence (Mitchell, 2014). Composite indices combine patient experience, process adherence, and safety events (Shwartz et al., 2015).

2.2 Risk Adjustment

Risk adjustment controls for patient severity to ensure performance differences reflect institutional quality (Iezzoni, 1997; Werner & Bradlow, 2006).

2.3 Predictive Modeling

Linear regression predicts continuous quality scores (Ash et al., 2012); logistic regression classifies facilities (Mukamel et al., 2008). BigQuery ML enables SQL-based model training (Google Cloud, 2023).

2.4 Multi-Tool Workflows

Integrated BigQuery, Python, R, and Looker pipelines improve reproducibility in healthcare analytics (Reis & Housley, 2022; Simpao et al., 2014).

3. Methodology

3.1 Data Source

Dataset: 120 California hospitals, 7 variables. All statistics computed via Python (pandas). Code at github.com/PatelSahil-HDS.

Table 1 Dataset Variables and Descriptive Statistics (n = 120)

Variable	Type	Range / Values	Mean	SD
Performance_Rating	Continuous	55.8 – 100.0	75.50	9.43
Risk_Adjusted_Rate	Continuous	1.00 – 8.97	5.58	1.46
Cases	Integer	253 – 346	295.4	18.23
Adverse_Events	Integer	0 – 14	6.08	2.97
Hospital_Type	Categorical	General 45%, Teaching 18.3%, Community 20%, Specialty 16.7%	—	—
Adverse_Rate*	Engineered	0.000 – 0.046	0.021	0.010
High_Performer*	Binary	1 if Rating $\geq$ 75 (55.8% High, 44.2% Low)	—	—

Note. * = engineered. Adverse_Rate = Adverse_Events / Cases.

3.2 Data Cleaning

Zero missing values, zero duplicates. Raw Adverse_Events and Cases excluded from regression to prevent multicollinearity; Adverse_Rate retained. One-hot encoded with Community (type) and East (region) as references.

3.3 EDA and Dashboards

EDA: Python (matplotlib, seaborn) and R (ggplot2, corrplot). Dashboards in Power BI Desktop and Tableau Desktop. Published Tableau workbook: public.tableau.com/app/profile/sahil.patel (replace with your actual Tableau Public URL). Looker Studio connects to BigQuery for live-refresh reporting.

3.4 Models

3.4.1 Linear Regression

Outcome: Performance_Rating. Features: Risk_Adjusted_Rate, Adverse_Rate, 6 binary indicators. All z-score normalised. Evaluated via R^2 , MAE, RMSE ($n_{\text{test}}=30$) and 5-fold CV. Also trained via BigQuery ML and R `lm()`.

3.4.2 Logistic Regression

Outcome: High_Performer (0/1). L2-regularised LogisticRegression ($C=1.0$, $\text{max_iter}=1000$). Evaluated via accuracy, AUC, precision, recall, F1. Also via R `glm()` and BigQuery ML.

3.5 Reproducibility

Python 3.10, R 4.3.0, BigQuery ML, Power BI, Tableau 2024.1, Looker Studio. All code: github.com/PatelSahil-HDS.

4. Results

4.1 Descriptive Findings

Mean Performance_Rating = 75.50 (SD = 9.43, median = 76.6, range: 55.8–100.0). General hospitals: n=54 (45.0%). 67 hospitals (55.8%) classified as High Performers. All Pearson $|r| \leq 0.20$.

Table 2 Mean Performance Rating by Hospital Type (Real Data)

Hospital Type	n	Mean	SD	Min	Max
General	54	76.63	9.11	61.0	100.0
Teaching	22	75.65	10.66	59.5	100.0
Community	24	74.95	9.43	58.9	96.9
Specialty	20	72.92	9.00	55.8	86.4

Table 3 Pearson Correlations (Real Data)

Variable	Perf. Rating	Risk Adj.	Cases	Adv. Events
Performance Rating	1.000	0.081	−0.029	−0.078
Risk Adjusted Rate	0.081	1.000	−0.049	−0.068
Cases	−0.029	−0.049	1.000	0.202
Adverse Events	−0.078	−0.068	0.202	1.000

4.2 Linear Regression

Test $R^2 = -0.24$, MAE = 7.96, RMSE = 9.99. CV $R^2 = -0.15$. Negative R^2 confirms the model underperforms a naive mean baseline. Region_North ($\beta = +1.46$) and Region_West ($\beta = +1.23$) were the largest positive coefficients; Adverse_Rate ($\beta = -1.07$) the largest negative.

Table 4 Linear Regression — Real Test Set Performance ($n_{test} = 30$)

Metric	Value	Interpretation
R^2 (Test)	−0.24	Worse than naive mean; weak feature set
MAE	7.96	Average absolute error in rating points
RMSE	9.99	Root mean squared error
CV R^2 (5-fold)	−0.15	Poor generalisation confirmed
Intercept	76.49	Rating at mean standardised features

4.3 Logistic Regression

Accuracy = 56.7%, AUC = 0.638. High recall for High Performers (0.92); low specificity for Low Performers (0.29). Confusion matrix: 5 TN, 12 FP, 1 FN, 12 TP.

Table 5 Logistic Regression — Classification Report ($n_{test} = 30$)

Class	Precision	Recall	F1	Support
Low Performer (0)	0.83	0.29	0.43	17
High Performer (1)	0.50	0.92	0.65	13
Macro Avg	0.67	0.61	0.54	30
Weighted Avg	0.69	0.57	0.53	30

Table 6 Logistic Regression — Odds Ratios (Real Data)

Feature	Coefficient	Odds Ratio	Direction
Region_West	+0.56	1.75	Strongest positive predictor
Hospital_Type_General	+0.33	1.39	General type → higher High Performer odds
Region_North	+0.29	1.34	North region positive
Adverse_Rate	−0.23	0.79	Higher adverse rate → lower odds
Hospital_Type_Teaching	−0.10	0.90	Slight negative

5. Discussion

5.1 Linear Regression: Honest Null Findings

The negative R^2 confirms empirically that composite quality scores are driven by factors beyond standard administrative variables (Shwartz et al., 2015): staffing ratios, EHR adoption, payer mix, and socioeconomic context were unavailable. The finding motivates richer data collection.

5.2 Logistic Regression

AUC = 0.638 demonstrates meaningful discrimination above chance. High recall for High Performers (0.92) makes it useful as a screening tool. Adverse_Rate (OR = 0.79) was the strongest negative predictor, consistent with prior literature (Needleman et al., 2011).

5.3 Multi-Tool Scalability

BigQuery ML enables real-time hospital scoring. Tableau dashboard: public.tableau.com/app/profile/sahil.patel. Looker Studio provides auto-refreshing web dashboards.

5.4 Limitations

Key limitations: synthetic data, narrow feature set, $n=120$, linear assumptions, arbitrary threshold of 75. Code archived at github.com/PatelSahil-HDS.

6. Conclusion

This study applied a multi-tool pipeline to 120 California hospitals. Linear $R^2 = -0.24$ is an honest empirical finding; logistic AUC = 0.638 shows modest discrimination. All code at github.com/PatelSahil-HDS.

Future work should incorporate staffing data, apply ensemble models, conduct longitudinal analysis, and validate on authenticated CMS datasets.

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